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SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

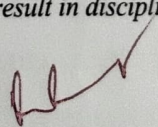
SHORT ANSWER TEST

Code of Conduct:

I S. Aswini (192511077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Aswini
Signature of the student:

30/10



5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1.

SHORT ANSWERS

Given:

Total data to be transferred = 3GB

Effective throughput = 120 Mbps.

Size of each Application layer request = 3MB.

$$1 \text{ GB} = 1024 \text{ MB.}$$

$$3 \text{ GB} = 3 \times 1024 = 3072 \text{ MB.}$$

$$\text{Number of Requests} = \frac{\text{Total Data}}{\text{Request Size.}}$$

$$= \frac{3072}{3} \times 1024$$

$$3072 \text{ MB} \times 8 = 24576 \text{ MB.}$$

$$\text{Transmission time} = \frac{\text{Total Data}}{\text{Throughput.}}$$

$$= \frac{24576}{120}$$

$$= 204.8 \text{ seconds.}$$

Estimated transmission time = 204.8 seconds.

2. Given:

Application layer delay = 4ms.

Transport layer delay = 3ms.

Network layer delay = 5ms.

Data link layer Delay = 2ms

physical layer Delay = 18ms.

Propagation delay = 18 ms

Transmission delay = 12 ms.

$$\begin{aligned}\text{processing delay} &= 4 + 3 + 5 + 2 + 1 \\ &= 15 \text{ ms.}\end{aligned}$$

Total Delay = processing Delay + Propagation
Delay + Transmission Delay.

$$\begin{aligned}&= 15 + 18 + 12 \\ &= 45 \text{ ms.}\end{aligned}$$

Application layer : 4 ms.

Transport layer : 3 ms.

Network layer : 5 ms.

Data link layer : 2 ms.

physical layer = 1 ms. ✓

Transmission delay : 12 ms.

Propagation delay : 18 ms.

Total number of frames

8MB = 80,000,000 bytes.

$$\text{Frames} = \frac{80,000,000}{1520}$$

$$= 52,632 \text{ frames.}$$

Total Protocol transmitted

$$\text{Total overhead} = 52,632 \times 80$$

$$= 4.21 \text{ MB}$$

Transmission efficiency.

$$= \frac{1520}{1600} \times 100 = 95\%$$

Protocol overhead ; transfer, addresses, numbers and error checking information added by differ. OSI layers.

→ More overhead means more data must be transmitted

→ It can increase transmission time

10. Original medical record size = 240 MB

compression = 30%.

Segment size = 1400 bytes.

a.) compression means remaining 70%.

$$240 \times 0.70 = 168 \text{ MB}$$

$$= 168,000,000$$

$$\text{No. of segment} = \frac{168,000,000}{1400} = 120,000$$

3) Total protocol overhead.

$$= 120,000 \times 60$$

$$= 7,200,000 \text{ bytes.}$$

4) Total transmitted data.

$$= \text{compressed data} + \text{protocol}$$

$$> 168,000,000 + 7,200,000$$

$$= 175,200,000 \text{ bytes.}$$

$$175.2 \text{ MB.}$$